

Breadth First Search (BFS)

Aim

To implement the **Breadth-First Search (BFS)** algorithm in Python to traverse all the vertices of a graph level by level.

Algorithm

1. Create a graph using an adjacency list.
2. Choose a starting node.
3. Mark the starting node as visited and insert it into a queue.
4. Remove a node from the front of the queue.
5. Visit all its unvisited adjacent nodes.
6. Mark them as visited and add them to the queue.
7. Repeat Steps 4–6 until the queue becomes empty.

Python Code

```
from collections import deque
```

```
# Input number of vertices
```

```
n = int(input("Enter the number of vertices: "))
```

```
graph = {}
```

```
# Input adjacency list
```

```
for i in range(n):
```

```
    vertex = input(f"Enter vertex {i+1}: ")
```

```
    neighbours = input(f"Enter neighbours of {vertex} (space-separated): ").split()
```

```
    graph[vertex] = neighbours
```

```
start = input("Enter the starting vertex: ")
```

```

def bfs(graph, start):
    visited = set()
    queue = deque([start])
    visited.add(start)

    print("\nBFS Traversal:")

    while queue:
        node = queue.popleft()
        print(node, end=" ")

        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)

bfs(graph, start)

```

Output

```

Enter the number of vertices: 5
Enter vertex 1: A
Enter neighbours of A (space-separated): B C
Enter vertex 2: B
Enter neighbours of B (space-separated): D
Enter vertex 3: C
Enter neighbours of C (space-separated): E
Enter vertex 4: D
Enter neighbours of D (space-separated):
Enter vertex 5: E
Enter neighbours of E (space-separated):
Enter the starting vertex: A

BFS Traversal:
A B C D E

```

Result

The **Breadth-First Search (BFS)** algorithm was successfully implemented in Python. The program traversed the graph level by level using a queue and visited all the vertices successfully.